

Introduction to Air Dispersion Modeling with PyAERMOD

A hands-on guide for students who have never used a dispersion model before. No programming is required — all five tutorials use the PyAERMOD graphical interface (GUI) in your web browser.

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1. What Is Air Dispersion Modeling?

When a factory, power plant, or vehicle emits pollutants into the air, those pollutants don't stay in one place. Wind carries them downwind, and turbulence spreads them out in all directions. An **air dispersion model** is a computer program that predicts *where* those pollutants go and *how concentrated* they are at ground level.

Why Does It Matter?

- **Protecting public health.** Regulatory agencies (like the U.S. EPA) use dispersion models to decide whether a proposed factory will cause air quality problems in nearby neighborhoods.
- **Designing better facilities.** Engineers use models to determine how tall a smokestack needs to be, or whether adding a scrubber will bring concentrations below safe limits.
- **Emergency planning.** If there's an accidental release of a toxic gas, models help predict which areas might be affected.

What Is AERMOD?

AERMOD (AMS/EPA Regulatory Model) is the U.S. EPA's preferred model for near-field (< 50 km) regulatory air quality assessments. It has been the standard since 2005 and is used worldwide.

AERMOD is a **steady-state Gaussian plume model**, which means:

- **Steady-state:** it assumes meteorological conditions are constant during each hour of simulation.
- **Gaussian plume:** it models the spreading pollutant cloud as a bell-shaped (Gaussian) distribution in both the horizontal and vertical directions.

PyAERMOD is a Python wrapper that lets you set up, run, and analyze AERMOD models through a graphical interface or Python code, instead of editing text files by hand.

2. Key Concepts

Before diving into the tutorials, here are the core ideas you'll need.

2.1 Emission Sources

An **emission source** is anything that releases pollutants into the air. AERMOD supports several types:

Source Type	Real-World Example
Point	Smokestack, exhaust vent
Area	Storage pile, parking lot, landfill
Volume	Conveyor transfer point, building vent
Line	Roadway, railway, pipeline
Open Pit	Quarry, surface mine

For these tutorials, we'll focus on **point sources** (smokestacks) because they're the most common and the easiest to understand.

Key Point Source Parameters

- **Stack height** (meters): How tall the smokestack is. Taller stacks release pollutants higher up, giving them more time to dilute before reaching the ground.
- **Stack diameter** (meters): The inner diameter of the stack opening.
- **Exit velocity** (m/s): How fast the exhaust gas leaves the stack. Faster gas shoots the plume higher (this is called **momentum rise**).
- **Exit temperature** (Kelvin): Hot exhaust gas is buoyant — it rises because it's lighter than the surrounding air. This is called **buoyancy rise**. The combination of momentum and buoyancy rise is called **plume rise**.
- **Emission rate** (g/s): The mass of pollutant released per second.

| **Intuition:** Imagine holding a lit candle. The hot smoke rises (buoyancy). If you blow on it, the smoke bends sideways (wind). If you raise the candle overhead, the smoke takes longer to reach the floor (stack height). These are the same physics AERMOD captures mathematically.

2.2 Receptors

Receptors are the locations where AERMOD calculates the ground-level pollutant concentration. Think of them as virtual air quality monitors placed across a map.

Common receptor layouts:

- **Cartesian grid:** A regular rectangular grid of points (like graph paper laid over a map). You specify the boundaries and spacing. A 2 km x 2 km area with 100 m spacing gives 441 receptors.
- **Polar grid:** Rings of points radiating out from a central location (like a dartboard). Useful for studying how concentration changes with distance from a source.
- **Discrete receptors:** Individual points at specific locations of interest (a school, a hospital, a property boundary).

2.3 Meteorology

Wind and atmospheric stability are the two most important factors controlling dispersion:

- **Wind speed and direction** determine where the plume goes.
- **Atmospheric stability** determines how much the plume spreads:
- **Unstable** (sunny daytime): strong vertical mixing, plume spreads rapidly, concentrations drop quickly with distance.
- **Stable** (clear nighttime): weak mixing, plume stays narrow and compact, can travel far with less dilution.
- **Neutral** (overcast, windy): moderate mixing.

AERMOD uses **hourly meteorological data** — typically one to five years of observations from a nearby weather station, preprocessed by the AERMET program into `.sfc` (surface) and `.pfl` (profile) files. For tutorials 1 and 2 you won't need actual met data because we'll only generate the input file. Tutorial 3 shows how to connect real meteorological files.

2.4 What AERMOD Produces

For each receptor, AERMOD calculates the **pollutant concentration** (typically in micrograms per cubic meter, $\mu\text{g}/\text{m}^3$) for each hour of the simulation. It then computes summary statistics:

- **Annual average** — the mean concentration over all hours. Compared to the annual NAAQS (National Ambient Air Quality Standard).
- **24-hour average** — the highest 24-hour average over the simulation period.
- **1-hour average** — the highest single-hour concentration.

The results let you answer questions like: *"Will this factory cause the annual PM_{2.5} concentration at the nearest school to exceed 12 $\mu\text{g}/\text{m}^3$?"*

2.5 The Modeling Workflow

Every AERMOD analysis follows this sequence:

```
Define the problem
  |
  v
Set up sources, receptors, and meteorology
  |
  v
Generate the AERMOD input file (.inp)
  |
  v
Run AERMOD --> Output file (.out)
  |
  v
Analyze results (tables, maps, compliance checks)
```

The PyAERMOD GUI walks you through each of these steps in order.

3. Setting Up PyAERMOD

3.1 Install

Open a terminal (Command Prompt on Windows, Terminal on Mac/Linux) and run:

```
pip install pyaermod[gui]
```

This installs the core package plus everything needed for the graphical interface (Streamlit, Folium maps, matplotlib plots).

To verify the install worked:

```
pyaermod
```

You should see version information and a feature list.

3.2 Install the AERMOD and AERMET Executables

PyAERMOD is a *wrapper* — it generates input files, launches the model, and visualizes results, but it does **not** include the AERMOD or AERMET programs themselves. Those are separate Fortran executables maintained by the U.S. EPA.

Which executables do you need?

Executable	What it does	Which tutorials need it
aermod	Runs the dispersion model	Tutorial 3 (running AERMOD)
aermet	Preprocesses meteorological data	Tutorial 5 (AERMET processing)
aermap	Preprocesses terrain elevations	Optional (terrain analysis)

Tutorials 1–2 only generate input files, so you can **skip this step initially** and come back when you're ready for Tutorial 3.

Option A: Download pre-compiled binaries (macOS Apple Silicon)

Pre-compiled macOS binaries for aermod and aermap are attached to each PyAERMOD release on GitHub.

1. Go to <https://github.com/atmmod/pyaermod/releases/latest>
2. Scroll down to the **Assets** section at the bottom of the page.
3. Click on aermod and aermap to download them.

| **Note:** Your browser may warn that these are "unverified" downloads. This is normal for command-line executables — click **Keep** or **Allow** to save the files.

4. Move them to a permanent location and make them executable:

```
mkdir -p ~/bin
mv ~/Downloads/aermod ~/Downloads/aermap ~/bin/
chmod +x ~/bin/aermod ~/bin/aermap
```

5. Add ~/bin to your PATH so PyAERMOD can find them. Add this line to your ~/.zshrc (or ~/.bashrc on Linux):

```
export PATH="$HOME/bin:$PATH"
```

Then **restart your terminal** (close and reopen it), or run `source ~/.zshrc` in your current session.

6. Verify they work:

```
which aermod  
aermod
```

The which command should print ~/bin/aermod (or the full expanded path). Running aermod alone will print a version header and then exit with an error about missing input files — **that's normal** and means it's installed correctly.

| **AERMET is not included in the release binaries.** If you need AERMET for Tutorial 5, use Option B below to compile it from source, or ask your instructor for a pre-compiled copy.

Option B: Compile from source (macOS Intel, Linux, or to get AERMET)

This option compiles **all three executables** (AERMOD, AERMAP, and AERMET) from the Fortran source code included in the PyAERMOD repository.

Prerequisites: You need gfortran (the GNU Fortran compiler).

On macOS, install it with Homebrew:

```
brew install gcc
```

On Ubuntu/Debian Linux:

```
sudo apt-get install gfortran
```

Build steps:

```
git clone https://github.com/atmmod/pyaermod.git  
cd pyaermod  
./scripts/build_aermod.sh all
```

This creates executables in the bin/ directory. Move them to your PATH:

```
mkdir -p ~/bin  
cp bin/aermod bin/aermap ~/bin/
```

If AERMET source is available in the aermet/ directory, the build script will compile it too. Copy it the same way:

```
cp bin/aermet ~/bin/
```

Then add ~/bin to your PATH as described in Option A, step 5 above.

Option C: Windows

The EPA provides official Windows executables:

1. Go to the EPA SCRAM website.
2. Download the **AERMOD Modeling System** zip file.
3. Extract the zip. Inside you'll find aermod.exe, aermet.exe, and aermap.exe.
4. Place them in a folder (e.g., C:\AERMOD\) and add that folder to your system PATH: - Search for "**Environment Variables**" in the Start menu. - Under **System variables**, find Path, click **Edit**. - Click **New** and add C:\AERMOD\ (or wherever you put the files). - Click **OK** to save.
5. Open a **new** Command Prompt and verify:

```
aermod
```

Verifying your installation

After installing by any method, you can verify everything is set up with:

```
pyaermod          # Should show PyAERMOD version and feature list
which aermod      # Should print the path to the aermod executable
aermod            # Should print AERMOD version (error about input is OK)
```

| The GUI's "Run AERMOD" page will automatically search your PATH for the executable. If it can't find it, you'll see an error message with instructions. You can also specify the full path to the executable directly in the GUI.

3.3 Launch the GUI

```
pyaermod-gui
```

Your web browser will open automatically to `http://localhost:8501`. You'll see the PyAERMOD interface with a sidebar on the left and the **Project Setup** page in the main area.

| **Tip:** The GUI runs as a local web app. It never sends your data to the internet — everything stays on your computer.

3.4 GUI Layout

- **Sidebar (left):** Navigation between the 7 workflow pages, plus a progress checklist showing which steps are complete.
- **Main area (center):** The active page — forms, maps, tables, and plots.

The 7 pages, in workflow order:

1. **Project Setup** — project name, pollutant, terrain, coordinate system
 2. **Source Editor** — add and configure emission sources on a map
 3. **Receptor Editor** — define where concentrations are calculated
 4. **Meteorology** — point to weather data files
 5. **Run AERMOD** — validate, preview, and execute the model
 6. **Results Viewer** — maps, plots, statistics, compliance checks
 7. **Export** — download results as GeoTIFF, Shapefile, CSV, etc.
-

4. Tutorial 1 — Your First Point Source

Goal: Create a simple AERMOD input file for a single smokestack emitting PM2.5, and understand what each setting means.

Time: 15--20 minutes

What you'll learn:

- How to configure a project in the GUI
- How to define a point source (smokestack)
- How to set up a receptor grid
- How to specify meteorology file paths
- How to preview and download the generated AERMOD input file

Step 1: Project Setup

1. Launch the GUI (pyaermod-gui).
2. On the **Project Setup** page, fill in: - **Title Line 1:** Tutorial 1 - My First Model - **Title Line 2:** Single point source, PM2.5 - **UTM Zone:** Pick the zone for your area. (If unsure, use 17 for the eastern U.S. or 10 for the western U.S. The exact zone doesn't matter for learning.) - **Hemisphere:** N (Northern) - **Datum:** WGS84 - **Map Center Latitude:** 35.0 (or your local latitude) - **Map Center Longitude:** -80.0 (or your local longitude) - **Pollutant Type:** PM25 - **Terrain Type:** FLAT - **Averaging Periods:** Select ANNUAL and 24

| **What you just did:** You told AERMOD what pollutant you're modeling (PM2.5), that the terrain is flat (no hills), and that you want annual and 24-hour average results. The coordinate system settings let the GUI convert between the map (latitude/longitude) and AERMOD's coordinate system (UTM meters).

Step 2: Add a Point Source

1. Click **Source Editor** in the sidebar.
2. You'll see an interactive map. Below it, select source type: **Point**.
3. Fill in the source form:

Parameter	Value	What It Means
Source ID	STACK1	A short name for this source
X Coordinate	500000	Easting in UTM meters (or click the map)
Y Coordinate	3870000	Northing in UTM meters (or click the map)
Base Elevation	0	Ground elevation at the stack base (meters)
Stack Height	50	The stack is 50 meters tall
Exit Temperature	400	Exhaust gas temperature: 400 K (about 127 C)
Exit Velocity	15	Gas exits at 15 m/s

Stack Diameter	2	Stack opening is 2 meters across
Emission Rate	1.5	Emitting 1.5 grams of PM2.5 per second

4. Click **Add Source**.

| **What you just did:** You defined a medium-sized industrial smokestack. The hot, fast-moving exhaust gas will rise well above the 50 m physical stack height (plume rise), so the "effective" release height might be 80--120 m depending on meteorological conditions. This is why AERMOD needs the temperature and velocity — not just the physical height.

You should see your source appear as a marker on the map and in the source table below.

Step 3: Set Up Receptors

1. Click **Receptor Editor** in the sidebar.
2. On the **Cartesian Grid** tab, enter:

Parameter	Value	What It Means
X Min	498000	Western edge of the grid
X Max	502000	Eastern edge (4 km total width)
Y Min	3868000	Southern edge
Y Max	3872000	Northern edge (4 km total height)
Spacing	200	One receptor every 200 meters

3. The GUI will show: **21 x 21 = 441 receptors**. Click **Add Grid**.

| **What you just did:** You placed a 4 km x 4 km grid of virtual air quality monitors centered on your source, with a receptor every 200 meters. AERMOD will calculate the ground-level PM2.5 concentration at each of these 441 points.

Step 4: Specify Meteorology

1. Click **Meteorology** in the sidebar.
2. Select **Use Existing Files** mode.
3. Enter file paths: - **Surface file:** met_data.sfc - **Profile file:** met_data.pfl

| **Note:** You don't need actual meteorological files for this tutorial — we're just generating the input file. When you're ready to actually run AERMOD (Tutorial 3), you'll need real .sfc and .pfl files.

Step 5: Preview the Input File

1. Click **Run AERMOD** in the sidebar.
2. Expand the **Generated Input File** section. You'll see the complete AERMOD input file — a text file organized into five pathways:
 - **CO** (Control): Project title, pollutant, options
 - **SO** (Source): Your stack parameters
 - **RE** (Receptor): Your grid definition

- **ME** (Meteorology): File paths
 - **OU** (Output): What results to produce
3. Click **Download Input File** to save it as `aermod.inp`.

| **What you just did:** You generated a complete, valid AERMOD input file without editing a single line of text. Open the `.inp` file in a text editor and compare it to what the GUI showed you — they're identical.

Checkpoint

At this point you should understand:

- ✓ The five AERMOD pathways (CO, SO, RE, ME, OU)
 - ✓ What stack height, temperature, velocity, and emission rate represent
 - ✓ How a Cartesian receptor grid covers an area around the source
 - ✓ That AERMOD needs hourly meteorological data in `.sfc / .pfl` format
-

5. Tutorial 2 — Comparing Stack Heights

Goal: Create two versions of the same facility — one with a 20 m stack and one with a 60 m stack — to understand how stack height affects ground-level concentrations.

Time: 15--20 minutes

What you'll learn:

- How stack height controls plume rise and ground-level impact
- How to use the Project Save/Load feature to create scenario variants
- How to read an AERMOD input file and spot key differences

Conceptual Background

Imagine two identical factories, both emitting 2 g/s of SO₂. The only difference is the smokestack:

- **Factory A:** 20 m stack (about 6 stories tall)
- **Factory B:** 60 m stack (about 20 stories tall)

Which factory will cause higher ground-level concentrations? Intuitively, Factory A — its emissions are released closer to the ground, so they have less distance to dilute before reaching people.

But *how much* of a difference does it make? That's exactly what AERMOD calculates.

Step 1: Build the Base Scenario

Follow the same steps as Tutorial 1, but with these settings:

Project Setup:

- Title: Tutorial 2 - Stack Height Comparison (20m)
- Pollutant: SO₂
- Averaging Periods: 1, 24, and ANNUAL
- Terrain: FLAT

Source (Point):

- Source ID: STACK1
- Stack Height: 20
- Exit Temperature: 420 K
- Exit Velocity: 15 m/s
- Stack Diameter: 2 m
- Emission Rate: 2.0 g/s

Receptors (Cartesian Grid):

- 4 km x 4 km area centered on the source, 100 m spacing

Meteorology:

- Surface file: met_data.sfc
- Profile file: met_data.pfl

Step 2: Save the Project

1. Go back to **Project Setup**.
2. Click **Download Project** to save the project as a JSON file. Name it `tutorial2_20m.json`.

Step 3: Download the 20 m Input File

1. Go to **Run AERMOD**.
2. Download the input file. Name it `stack_20m.inp`.

Step 4: Create the 60 m Variant

1. Go to **Source Editor**.
2. Delete the existing source (select it in the table and click **Delete**).
3. Add a new source with the same parameters, **except Stack Height: 60**.
4. Update the project title to `Tutorial 2 - Stack Height Comparison (60m)`.
5. Save this project as `tutorial2_60m.json`.
6. Download the input file as `stack_60m.inp`.

Step 5: Compare the Input Files

Open both `.inp` files in a text editor and find the `S0 SRCPARAM` line:

20 m stack:

S0	SRCPARAM	STACK1	2.0	20.0	420.0	15.0	2.0
----	----------	--------	-----	------	-------	------	-----

60 m stack:

S0	SRCPARAM	STACK1	2.0	60.0	420.0	15.0	2.0
----	----------	--------	-----	------	-------	------	-----

The only difference is the second number (stack height). Everything else is identical, making this a controlled experiment.

What to Expect When You Run These (Tutorial 3)

When you eventually run both files through AERMOD with real meteorological data:

- The **20 m stack** will produce **higher** maximum ground-level concentrations, especially close to the source (within 500 m).
- The **60 m stack** will produce **lower** maximum concentrations because the plume is released higher and has more room to dilute.
- The **location** of the maximum concentration will also shift — taller stacks push the peak concentration **farther downwind** because the plume takes longer to mix down to ground level.

This is a fundamental principle in air quality engineering: **effective stack height is the single most important factor for ground-level concentrations** from a point source.

Checkpoint

At this point you should understand:

- ✓ Taller stacks reduce ground-level concentrations

- ✓ Taller stacks push the peak concentration farther downwind
 - ✓ The Project Save/Load feature lets you create scenario variants
 - ✓ The SO SRCPARAM line contains all point source parameters
-

6. Tutorial 3 — Running AERMOD and Reading Results

Goal: Run AERMOD with real meteorological data and interpret the results using the GUI's Results Viewer.

Time: 30--45 minutes

Prerequisites:

- The aermod executable must be installed and on your PATH. If you haven't done this yet, go back to **Section 3.2** and follow the instructions for your operating system.
- Meteorological data files (.sfc and .pfl). Your instructor may provide these, or you can process your own with AERMET (Tutorial 5).

| **If you don't have met data yet:** Your instructor should provide a pair of .sfc and .pfl files for your region. These are typically generated from NOAA weather station data using the AERMET preprocessor. Ask your instructor which files to use and where to find them. **If AERMOD is not installed:** Run which aermod (Mac/Linux) or where aermod (Windows) in a terminal. If nothing is printed, go back to Section 3.2 to install it.

Step 1: Verify AERMOD Is Installed

Open a terminal and type:

```
aermod
```

If AERMOD is installed, you'll see an error about missing input files (that's OK — it just confirms the executable is found). If you get "command not found," you need to install AERMOD or add it to your PATH.

Step 2: Set Up the Model

Load one of your Tutorial 2 project files, or create a new project:

Project Setup:

- Title: Tutorial 3 - Full AERMOD Run
- Pollutant: PM25
- Averaging Periods: ANNUAL and 24
- Terrain: FLAT

Source:

- A single point source with realistic parameters (see Tutorial 1)

Receptors:

- Cartesian grid: 4 km x 4 km, 100 m spacing (441 receptors)

Meteorology:

- Click **Meteorology** in the sidebar.
- Select **Use Existing Files**.
- Enter the paths to your .sfc and .pfl files. You can also use the **Upload** buttons to copy the files into the working directory.

Step 3: Validate and Run

1. Click **Run AERMOD** in the sidebar.
2. The GUI will run validation checks. Fix any errors that appear (common issues: missing met file paths, no sources defined, no receptors defined).
3. Verify the **Working Directory** path — this is where AERMOD will write its output.
4. Verify the **AERMOD Executable Path** (default: aermod).
5. Click **Run AERMOD**.

A spinner will appear while AERMOD runs. Depending on your grid size and the number of hours of met data, this may take anywhere from 10 seconds to a few minutes.

When it finishes, you'll see either:

- A green **success** banner with runtime info, or
- A red **error** banner. Read the error message — the most common issue is AERMOD not finding the meteorological files.

Step 4: Explore the Results

Click **Results Viewer** in the sidebar. You now have four tabs:

Tab 1: Interactive Map

- Select an averaging period (e.g., ANNUAL).
- The map shows color-coded concentration values at each receptor location, with your source marked.
- **Zoom in** to see individual receptor values. **Pan** to explore the spatial pattern.
- Notice the **plume shape**: concentrations are highest downwind of the source and decrease with distance. The elongated pattern reflects the prevailing wind direction in your met data.

Tab 2: Static Plots

- A contour plot showing smooth concentration isopleths (lines of equal concentration).
- The concentric pattern around the source shows how concentrations decrease with distance — the **concentration gradient**.

Tab 3: Statistics

This is where you do compliance analysis:

- **Maximum concentration**: The highest value at any receptor. This is what regulators compare to the NAAQS.
- **Mean and median**: How polluted the area is on average.
- **Top 10 receptors**: Where are the worst-case locations?
- **Exceedance analysis**: Enter the NAAQS value to see how many receptors exceed the standard.

Common NAAQS values to try:

Pollutant	Period	Standard (ug/m3)
PM2.5	Annual	9.0
PM2.5	24-hour	35

PM10	24-hour	150
SO2	1-hour	196
NO2	1-hour	188
CO	8-hour	10,000

Tab 4: POSTFILE Viewer (Advanced)

If you enabled POSTFILE output on the Run page, you can upload the file here to see hour-by-hour concentration animations. This is optional for beginners.

Step 5: Interpret What You See

Ask yourself these questions:

- 1. Where is the maximum concentration?** It should be downwind of the source, at a distance that depends on stack height and meteorology. For a 50 m stack, the peak is typically 500--2000 m downwind.
- 2. What direction does the plume extend?** This reflects the prevailing wind direction in your met data. If the wind usually blows from west to east, the highest concentrations will be east of the source.
- 3. Does the maximum exceed the NAAQS?** Enter the relevant standard in the exceedance analysis. If any receptors exceed it, the facility would need additional controls (taller stack, emission reduction, scrubbers, etc.).
- 4. How quickly do concentrations drop with distance?** Look at the contour plot — concentrations typically drop to 10% of the maximum within 1--2 km for elevated point sources.

Checkpoint

At this point you should understand:

- ✓ How to run AERMOD from the GUI
- ✓ How to read the interactive map and identify the plume direction
- ✓ How to find the maximum concentration and where it occurs
- ✓ How to check whether a facility complies with air quality standards
- ✓ That wind direction and stack height are the dominant factors in where the plume goes and how concentrated it is

7. Tutorial 4 — Area Sources: Modeling a Facility with Fugitive Emissions

Goal: Model a construction site with three types of area sources — a rectangular stockpile, a circular staging area, and an irregular polygon site boundary — alongside a point source (generator exhaust).

Time: 25--30 minutes

What you'll learn:

- The difference between point source and area source emission rates
- How to add rectangular, circular, and polygonal area sources in the GUI
- How to combine point and area sources in one model
- How source groups let you analyze contributions separately

Conceptual Background

Not all emissions come from smokestacks. Many facilities also have **fugitive emissions** — pollutants that escape from ground-level or low-height sources rather than through a defined stack. Examples include:

Source	Why It Emits
Dirt stockpile	Wind blows dust off exposed surfaces
Parking lot	Vehicle exhaust, tire/brake wear, re-entrained road dust
Construction site	Earth-moving equipment, exposed soil
Tank farm	Vapors escape through tank vents and seals
Landfill	Gases seep through the surface over a large area

These sources are modeled as **area sources** in AERMOD. The key difference from point sources is how emission rates are specified:

- **Point source:** total mass emitted per second (**g/s**)
- **Area source:** mass emitted per second *per unit area* (**g/s/m²**)

Because area sources spread emissions over a large surface, the per-square-meter rate is typically a very small number (e.g., 0.0001 g/s/m²).

| **Example calculation:** A 100 m x 50 m stockpile emits 0.5 g/s total. Area = 100 x 50 = 5,000 m².
Emission rate = 0.5 / 5,000 = **0.0001 g/s/m²**.

Step 1: Project Setup

1. Launch the GUI (pyaermod-gui).
2. On the **Project Setup** page: - **Title Line 1:** Tutorial 4 - Area Source Facility - **Title Line 2:** Construction site with mixed source types - **Pollutant Type:** PM₁₀ - **Averaging Periods:** Select 24 and ANNUAL - **Terrain Type:** FLAT

Step 2: Add a Point Source (Generator)

1. Click **Source Editor** in the sidebar.

2. Select source type: **Point**.

3. Fill in the form:

Parameter	Value	What It Represents
Source ID	GENSET	Diesel generator exhaust
X Coordinate	500050	UTM meters (or click the map)
Y Coordinate	3870050	UTM meters (or click the map)
Base Elevation	0	
Stack Height	5	Short exhaust stack
Exit Temperature	700	Diesel exhaust is very hot (K)
Exit Velocity	10	
Stack Diameter	0.3	Small pipe
Emission Rate	0.3	0.3 g/s of PM10

4. Click **Add Source**.

| **Why start with the point source?** It gives you a reference marker on the map. You'll place the area sources around it.

Step 3: Add a Rectangular Area Source (Stockpile)

1. In the source type selector, choose **Area (Rectangular)**.

2. Fill in the form:

Parameter	Value	What It Represents
Source ID	PILE1	Dirt/gravel stockpile
X Coordinate	500000	Southwest corner X
Y Coordinate	3870000	Southwest corner Y
Base Elevation	0	
Release Height	2.0	Dust lifts off ~2 m above the pile surface
Half-Width Y	25.0	50 m total in the Y direction
Half-Width X	50.0	100 m total in the X direction
Rotation Angle	0	Aligned with the grid (no rotation)
Emission Rate	0.000100	0.0001 g/s/m2 (see calculation above)

3. Click **Add Area Source**.

| **Understanding half-widths:** AERMOD defines rectangular area sources by their *half-widths* from the source coordinate. A "Half-Width X" of 50 means the source extends 50 m from the center in each X direction (100 m total width). This can be confusing at first — just remember to enter **half** the actual dimension.

Step 4: Add a Circular Area Source (Staging Area)

1. Select source type: **Area (Circular)**.
2. Fill in the form:

Parameter	Value	What It Represents
Source ID	STAGING	Equipment staging / laydown area
X Coordinate	500200	Center of the circle
Y Coordinate	3870100	
Base Elevation	0	
Release Height	1.0	Low-level dust from vehicle traffic
Radius	60	60 m radius circle
Num Vertices	20	How many sides to approximate the circle
Emission Rate	0.000050	Lower rate — less disturbed surface

3. Click **Add Circular Area Source**.

| **Num Vertices:** AERMOD approximates circles as polygons. 20 vertices gives a smooth-enough circle for most purposes. More vertices is more accurate but takes slightly longer to compute.

Step 5: Add a Polygonal Area Source (Site Boundary)

1. Select source type: **Area (Polygon)**.
2. Set **Number of Vertices** to 5 (this selector appears *above* the form).
3. Fill in the form:

Parameter	Value
Source ID	SITEBND
Base Elevation	0
Release Height	0.5
Emission Rate	0.000020

4. Enter vertex coordinates (these define the irregular site boundary):

Vertex	X	Y
V1	499900	3869900
V2	500350	3869900
V3	500400	3870100
V4	500300	3870250
V5	499900	3870200

5. Click **Add Polygon Source**.

| **Vertex order matters:** The vertices should trace the outline of the area in order (clockwise or counterclockwise). AERMOD closes the polygon automatically — you don't need to repeat the first vertex.

Step 6: Set Up Receptors

1. Click **Receptor Editor** in the sidebar.
2. On the **Cartesian Grid** tab:

Parameter	Value
X Min	499500
X Max	500800
Y Min	3869500
Y Max	3870700
Spacing	50

3. Click **Add Grid**. You should get roughly 600+ receptors covering the site and a buffer zone around it.

Step 7: Meteorology and Preview

1. Click **Meteorology > Use Existing Files >** enter paths to .sfc and .pfl files (or placeholder names for now).
2. Click **Run AERMOD** to preview the generated input file.

Expand the **Generated Input File** section and look for the SO pathway. You should see four sources — one POINT and three area types:

SO LOCATION	GENSET	POINT	500050.00	3870050.00	0.00
SO LOCATION	PILE1	AREA	500000.00	3870000.00	0.00
SO LOCATION	STAGING	AREAPOL	500200.00	3870100.00	0.00
SO LOCATION	SITEBND	AREAPOL	499900.00	3869900.00	0.00

Notice how each source type generates different SO SRCPARAM lines:

- **POINT:** emission rate, stack height, temperature, velocity, diameter
- **AREA:** emission rate, release height, half-width-Y, half-width-X
- **AREAPOL:** emission rate, release height, number of vertices

Step 8: Interpret Area Source Results (After Running)

When you eventually run this model, look for these patterns in the Results Viewer:

- **The generator (point source)** produces a concentrated plume downwind of the stack — a narrow elongated shape.
- **The stockpile (rectangular area)** produces a broader, lower-concentration pattern centered on the pile. Concentrations are highest at the downwind edge of the pile.
- **The site boundary (polygon)** produces the most diffuse pattern — low concentrations spread over a wide area.
- **Combined impact** is highest where plumes from multiple sources overlap.

Checkpoint

At this point you should understand:

- ✓ Area source emission rates are per unit area (g/s/m²), not total (g/s)
 - ✓ Rectangular area sources are defined by half-widths, not full dimensions
 - ✓ Circular sources are approximated by polygons (Num Vertices)
 - ✓ Polygon sources are defined by a list of corner coordinates
 - ✓ Different source types produce different spatial patterns in results
-

8. Tutorial 5 — Processing Meteorological Data with AERMET

Goal: Use the GUI's AERMET configuration page to generate the three-stage AERMET input files needed to process raw weather station data into AERMOD-ready meteorological files.

Time: 30--40 minutes

What you'll learn:

- What AERMET does and why AERMOD needs it
- How the three AERMET stages work
- How to configure each stage in the GUI
- What the monthly surface parameters mean and how to choose values

Prerequisites:

- The GUI generates AERMET *input files*, but to actually process them you need the `aermet` executable installed on your computer.
- **macOS/Linux:** If you followed Section 3.2, Option B (compile from source), you already have it. If you used Option A (download binaries), note that the release binaries include `aermod` and `aermap` but **not** `aermet` — go back and use Option B to compile it.
- **Windows:** The EPA's AERMOD Modeling System download from the SCRAM website includes `aermet.exe` alongside `aermod.exe`.
- Verify it's installed: run `which aermet` (Mac/Linux) or `where aermet` (Windows) in a terminal.

Conceptual Background

AERMOD cannot use raw weather data directly. Raw data from weather stations comes in formats designed for archival and general forecasting — not atmospheric dispersion modeling. AERMOD needs specially processed files that contain:

- **Hourly surface data** (`.sfc` file): wind speed, wind direction, ambient temperature, atmospheric stability class, mixing height, friction velocity, Monin-Obukhov length, and other boundary-layer parameters.
- **Hourly profile data** (`.pfl` file): vertical profiles of wind speed, wind direction, and temperature at multiple heights above ground.

AERMET is the preprocessor that transforms raw weather station observations into these two files. It runs in three stages:

```
Raw weather data (NOAA archives)
|
Stage 1: Extract & QA/QC
|
Extracted hourly data
|
Stage 2: Merge surface + upper air
|
Merged dataset
|
Stage 3: Compute boundary-layer parameters
|
aermod.sfc + aermod.pfl (ready for AERMOD)
```

What Data Does AERMET Need?

AERMET requires two types of weather observations:

1. **Surface observations** — hourly weather data from a ground-level station (typically an airport). Includes wind speed, wind direction, temperature, cloud cover, and ceiling height. In the U.S., this data is available from NOAA's Integrated Surface Hourly Data (ISHD) archive.
2. **Upper air soundings** — twice-daily vertical profiles of temperature, humidity, and wind measured by weather balloons (radiosondes). Available from NOAA's Radiosonde Database. The closest upper air station may be 100+ km away — that's normal.

| **Your instructor should provide** the station IDs, data files, and date ranges for your region. If you need to download your own data, NOAA's archives are at <https://www.ncei.noaa.gov/>.

Step 1: Navigate to the AERMET Page

1. Launch the GUI (pyaermod-gui).
2. First, set up your **Project Setup** page with the correct UTM zone and map center for your area (see Tutorial 1). AERMET Stage 3 may use these coordinates.
3. Click **Meteorology** in the sidebar.
4. Select the **Configure AERMET** radio button (not "Use existing files").

You'll see three tabs: **Stage 1**, **Stage 2**, and **Stage 3**.

Step 2: Configure Stage 1 — Extract & QA/QC

Click the **Stage 1: Data Extract** tab.

Purpose: Stage 1 reads raw weather station data, performs quality assurance checks (flagging missing or suspicious values), and extracts the variables AERMET needs.

Surface Station

Fill in the surface station information. Your instructor should provide these values. Example for Atlanta, GA:

Parameter	Example Value	What It Means
Station ID	KATL	ICAO airport code or WBAN number
Station Name	Atlanta Hartsfield	Descriptive name (for your reference)
Latitude	33.6300	Station latitude (decimal degrees, negative = south)
Longitude	-84.4400	Station longitude (decimal degrees, negative = west)
Time Zone (UTC offset)	-5	Eastern Standard Time = UTC-5
Elevation (m)	315.0	Station elevation above sea level
Anemometer Height (m)	10.0	Height of the wind sensor (usually 10 m)
Data Format	ISHD	NOAA Integrated Surface Hourly Data (most common)

| **Data Format options:** - **ISHD** — Integrated Surface Hourly Data (NOAA, post-2006). This is the most common format for recent U.S. data. - **HUSWO** — Hourly US Weather Observations (legacy NCDC format). - **SCRAM** — EPA SCRAM format (older pre-processed data). - **SAMSON** — Solar and Meteorological Surface Observational Network.

Upper Air Station

Fill in the upper air (radiosonde) station. Example:

Parameter	Example Value	What It Means
Station ID	72215	WMO station number
Station Name	Peachtree City	Closest radiosonde launch site
Latitude	33.3600	
Longitude	-84.5700	

| **Finding your upper air station:** The closest radiosonde station may be far from your surface station. In the U.S., there are only about 90 upper air stations (compared to thousands of surface stations). Use the one that is most representative of your area's upper-atmosphere conditions.

Data Files and Date Range

Parameter	Example Value	What It Means
Surface Data File	72219013874.dat	Path to the raw ISHD surface data file
Upper Air Data File	72215.dat	Path to the raw upper air data file
Start Date	2020/01/01	First day of the period to process
End Date	2020/12/31	Last day (typically 1--5 years of data)

Click **Save Stage 1 Configuration**.

Preview and Download

Expand **Preview Stage 1 Input** to see the generated AERMET input file. It will contain keywords like:

- JOB — identifies the processing job
- SURFACE DATA — points to the surface data file and specifies the format
- UPPERAIR DATA — points to the upper air data file
- XDATES — specifies the date range to extract

Click **Download Stage 1 Input File** to save it as aermet_stage1.inp.

Step 3: Configure Stage 2 — Merge

Click the **Stage 2: Merge** tab.

Purpose: Stage 2 takes the extracted surface and upper air data from Stage 1 and merges them into a single hourly dataset, aligning timestamps and interpolating upper air soundings to each hour.

Parameter	Example Value	What It Means
-----------	---------------	---------------

Surface Extract File	stage1.ext	Output from Stage 1 (surface data)
Upper Air Extract File	stage1_ua.ext	Output from Stage 1 (upper air)
Start Date	2020/01/01	Should match Stage 1
End Date	2020/12/31	Should match Stage 1
Merge Output File	stage2.mrg	Where to write the merged dataset

Click **Save Stage 2 Configuration**, then preview and download.

Step 4: Configure Stage 3 — Boundary Layer Parameters

Click the **Stage 3: Boundary Layer** tab.

Purpose: This is the most important stage. Stage 3 reads the merged data and computes the **planetary boundary layer parameters** that AERMOD needs: friction velocity, Monin-Obukhov length, convective velocity scale, mixing height, and more. These parameters depend on both the meteorological data and the **surface characteristics** around the station.

File Paths

Parameter	Example Value
Merge File	stage2.mrg
Start Date	2020/01/01
End Date	2020/12/31
Surface Output (.sfc)	aermod.sfc
Profile Output (.pfl)	aermod.pfl

Monthly Surface Parameters

This is the part that requires the most judgment. The GUI shows an editable table with three parameters for each month:

Parameter	What It Controls	Typical Range
Albedo	Surface reflectivity (fraction of sunlight reflected). Snow-covered ground has high albedo; dark pavement has low albedo.	0.10 -- 0.60
Bowen Ratio	Ratio of sensible heat to latent heat. Dry surfaces (deserts, cities) have high Bowen ratios; moist surfaces (wetlands, irrigated fields) have low values.	0.1 -- 10.0
Roughness (m)	Aerodynamic roughness length — how "bumpy" the surface is to wind flow. Open water is very smooth (~0.001 m); a city center is very rough (~1.0 m).	0.001 -- 2.0

The GUI pre-fills **suburban defaults** — these are reasonable starting values for an area with moderate development (residential neighborhoods, scattered trees):

Month	Albedo	Bowen Ratio	Roughness (m)
Jan	0.35	1.5	0.30
Feb	0.35	1.5	0.30
Mar	0.25	1.0	0.30
Apr	0.18	0.8	0.30
May	0.15	0.6	0.50
Jun	0.15	0.5	0.50
Jul	0.15	0.5	0.50
Aug	0.15	0.5	0.50
Sep	0.18	0.6	0.50
Oct	0.25	0.8	0.30
Nov	0.35	1.0	0.30
Dec	0.35	1.5	0.30

| **How to customize these values:** - **If your site is in a rural/agricultural area:** Lower roughness (0.05--0.15 m), albedo varies by crop/season, Bowen ratio varies by irrigation. - **If your site is in a dense urban area:** Higher roughness (0.5--1.5 m), higher Bowen ratio (dry impervious surfaces), lower albedo (dark pavement). - **If there's winter snow cover:** Increase albedo to 0.50--0.70 during snowy months. - **When in doubt:** Use the pre-filled suburban defaults. They're conservative and widely used in regulatory work.

Click in any cell in the table to edit it. The values update in real time.

Site Location

The **Use Stage 1 surface station** checkbox (checked by default) tells AERMET to use the latitude and longitude you entered in Stage 1. If you uncheck it, AERMET will use the project center coordinates from the Project Setup page.

Click **Save Stage 3 Configuration**.

Step 5: Download All Three Input Files

You now have three AERMET input files:

File	Stage	What It Does
aermet_stage1.inp	Extract & QA/QC	Reads raw data, checks quality
aermet_stage2.inp	Merge	Combines surface + upper air
aermet_stage3.inp	Boundary Layer	Computes AERMOD-ready parameters

Download all three from their respective preview sections.

Step 6: Run AERMET (Outside the GUI)

AERMET itself must be run separately (it's a different executable from AERMOD). First, verify that aermet is installed:

```
which aermet      # Mac/Linux – should print a path like ~/bin/aermet
where aermet      # Windows – should print a path like C:\AERMOD\ aermet.exe
```

If nothing is printed, go back to Section 3.2 to install it (Option B for Mac/Linux, Option C for Windows).

Then, in a terminal, navigate to the directory containing your downloaded input files and run each stage:

```
# Stage 1
aermet < aermet_stage1.inp

# Stage 2
aermet < aermet_stage2.inp

# Stage 3
aermet < aermet_stage3.inp
```

Each stage reads the output of the previous stage, so they must be run **in order**. When Stage 3 completes, you'll have your aermod.sfc and aermod.pfl files — the meteorological inputs AERMOD needs.

| **Troubleshooting common errors:** - *"File not found"*: Check that the data file paths in Stage 1 are correct and the files exist in the working directory. - *"Missing data exceeds threshold"*: Your raw data has too many gaps. AERMET flags this in the Stage 1 output. You may need to use a different station or a year with more complete records. - *"Upper air sounding missing"*: The upper air station may not have data for all dates. A few missing soundings are normal; AERMET interpolates.

Step 7: Use the Processed Met Data in AERMOD

Once you have aermod.sfc and aermod.pfl:

1. Go back to the **Meteorology** page in the GUI.
2. Switch to **Use existing .sfc/.pfl files** mode.
3. Enter the paths to your new aermod.sfc and aermod.pfl files.
4. Continue with your AERMOD run (Tutorial 3).

Alternatively, Stage 3's **Save** button automatically updates the Meteorology Pathway to point to the .sfc and .pfl files you specified, so you may already be set.

Understanding What AERMET Produced

The .sfc file contains one line per hour with columns including:

- Year, month, day, hour
- Sensible heat flux (W/m²) — energy driving vertical mixing
- Friction velocity (m/s) — a measure of wind turbulence near the ground
- Monin-Obukhov length (m) — characterizes atmospheric stability (negative = unstable, positive = stable)
- Mixing height (m) — the depth of the turbulent boundary layer
- Wind speed, direction, and temperature

The `.pfl` file contains vertical wind and temperature profiles for each hour — the information AERMOD uses to model how the plume disperses at different heights.

You don't need to read these files directly — AERMOD reads them automatically — but understanding what's in them helps you interpret your results. For example, if your maximum concentrations all occur during nighttime hours, it's likely because stable conditions (high Monin-Obukhov length, low mixing height) trapped the plume near the ground.

Checkpoint

At this point you should understand:

- ✓ Why AERMOD needs preprocessed meteorological data (not raw observations)
- ✓ The three AERMET stages: Extract, Merge, Boundary Layer
- ✓ That surface characteristics (albedo, Bowen ratio, roughness) vary by

land use and season

- ✓ How to generate the three AERMET input files using the GUI
- ✓ That AERMET must be run in order (Stage 1 → 2 → 3) to produce `.sfc`

and `.pfl` files

9. What's Next?

Now that you've completed all five tutorials, here are some directions to explore:

Advanced Tutorials: Houston Refinery Modeling (Tutorials 6--8)

The **Houston Refinery Modeling Assignments** extend this guide with three advanced tutorials that take you through a realistic industrial modeling project:

- **Tutorial 6** — Process meteorological data with AERMET for a specific Gulf Coast location (Houston Ship Channel), with site-appropriate surface parameters for a coastal/industrial area.
- **Tutorial 7** — Use AERMAP to process digital elevation data and assign terrain elevations to all receptors and sources, even in a relatively flat area like Houston.
- **Tutorial 8** — Build a complete AERMOD model for a simplified 150,000 barrel-per-day petroleum refinery with 10 emission sources (point, area, and volume), multi-scale receptors, source groups, regulatory compliance analysis, and sensitivity studies. Includes a technical report assignment.

These tutorials build directly on the skills from Tutorials 1--5 and include assignment deliverables and a grading rubric suitable for a university course.

Multiple Sources and Source Groups

Add several stacks and area sources to the same project and see how their plumes overlap. Use **source groups** (Source Editor > Source Groups) to separate contributions from different parts of a facility — for example, "STACKS" vs. "FUGITIVE" — and compare their relative impact.

Terrain Effects

Change the terrain type from FLAT to ELEVATED. Elevated terrain can significantly increase concentrations on hilltops that are close to the effective plume height. This requires terrain elevation data processed by AERMAP.

Background Concentrations

Real-world air quality is never zero — there's always some background pollution from regional sources, traffic, and natural sources. AERMOD can add a background concentration to its predictions using the **Background** option in the Source Editor (see Notebook 07 for the Python API approach).

The Python API

Once you're comfortable with the GUI concepts, try the Jupyter notebooks in `examples/notebooks/` to do the same work with Python code. The Python API gives you more flexibility for parameter sweeps, batch runs, and custom analysis.

Useful References

- AERMOD User's Guide (EPA) — The official 300+ page reference for all AERMOD options.
- AERMOD Model Formulation Document (EPA) — The mathematical basis behind the model.
- AERMET User's Guide (EPA) — Detailed documentation of all AERMET options and data formats.

- 40 CFR Part 51, Appendix W — EPA's Guideline on Air Quality Models (the regulatory framework for when and how to use AERMOD).
-

10. Glossary

Term	Definition
AERMOD	AMS/EPA Regulatory Model — the U.S. EPA's preferred near-field dispersion model
AERMET	AERMOD's meteorological preprocessor; converts raw weather data into .sfc and .pfl files
AERMAP	AERMOD's terrain preprocessor; extracts elevation data for receptors and sources
Averaging period	The time window over which concentrations are averaged (1-hour, 24-hour, annual, etc.)
Base elevation	The ground-level elevation at the base of a source or receptor (meters above sea level)
Buoyancy rise	The upward motion of a hot plume due to its lower density relative to ambient air
Cartesian grid	A rectangular array of evenly spaced receptor points
Concentration	The amount of pollutant in a given volume of air, typically in ug/m3
Discrete receptor	A single receptor placed at a specific location of interest
Effective stack height	Physical stack height + plume rise; the height at which the plume levels off
Emission rate	The mass of pollutant released per unit time (g/s for point sources, g/s/m2 for area sources)
Exceedance	When a predicted concentration is higher than the applicable air quality standard
Exit velocity	Speed of exhaust gas leaving the stack (m/s)
Gaussian plume	A mathematical model where pollutant concentration follows a bell curve in both horizontal and vertical cross-sections
Momentum rise	The upward motion of a plume due to the velocity of the exhaust gas
NAAQS	National Ambient Air Quality Standards — EPA's health-based concentration limits
Pathway	One of AERMOD's five input sections: CO (Control), SO (Source), RE (Receptor), ME (Meteorology), OU (Output)
Plume rise	The total height gain of exhaust gas above the physical stack top, from both buoyancy and momentum
Polar grid	A set of receptor points arranged in concentric rings at regular angular intervals around a center point
POSTFILE	An optional AERMOD output file containing concentration values at every receptor for every hour
Receptor	A location where AERMOD calculates ground-level pollutant concentration

Source group	A named subset of emission sources whose combined impact is reported separately
Stability	A measure of atmospheric turbulence: unstable (strong mixing), neutral, or stable (weak mixing)
Steady-state	An assumption that conditions (wind, stability) are constant during each simulation hour
Albedo	The fraction of incoming sunlight reflected by the ground surface (0 = absorbs all, 1 = reflects all). Fresh snow ~0.7, dark pavement ~0.1
Bowen ratio	Ratio of sensible heat flux to latent heat flux at the surface. High values (dry/urban), low values (moist/vegetated)
Friction velocity	A measure of wind-driven turbulence near the ground surface (m/s). Higher values mean more mechanical mixing
Fugitive emissions	Pollutants released from diffuse, ground-level sources (piles, lots, open areas) rather than through defined stacks
Half-width	AERMOD defines rectangular area sources by half the dimension in each direction from the source coordinate
ISHD	Integrated Surface Hourly Data — NOAA's standard format for recent U.S. surface weather observations
Mixing height	The depth of the atmospheric boundary layer (meters). Pollutants released below this height mix vertically within it
Monin-Obukhov length	A parameter characterizing atmospheric stability. Negative = unstable (good mixing), positive = stable (poor mixing)
Radiosonde	An instrument carried aloft by a weather balloon to measure vertical profiles of temperature, humidity, and wind
Roughness length	Aerodynamic parameter describing how rough the ground surface is to wind flow. Open water ~0.001 m, city center ~1.0 m
Surface characteristics	Albedo, Bowen ratio, and roughness length — the three land-surface parameters AERMET needs for each month
UTM	Universal Transverse Mercator — a coordinate system that measures position in meters (easting, northing) within numbered zones